INCH-POUND

MIL-PRF-19500/738C
17 April 2020
SUPERSEDING
MIL-PRF-19500/738B
24 January 2014

The documentation and process conversion measures necessary to comply with this revision shall be completed by 17 July 2020.

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, POWER DARLINGTON
TYPES 2N7575, 2N7576, AND 2N7577,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, power darlington transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package style is a three terminal flange mount header configuration (modified TO-254AA) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25 °C.

Types	P_T		$R_{\theta JC}$ (2)	V_{CBO} and V_{CEX}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}
	T_A = +25 °C	T_C = +25 °C (1)							
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	°C
2N7575	6	175	1.0	350	300	4.0	2.5	20	–65 to +200
2N7576	6	175	1.0	400	350	4.0	2.5	20	–65 to +200
2N7577	6	175	1.0	450	400	4.0	2.5	20	–65 to +200

(1) See figure 2 for temperature-power derating curves.
(2) See figure 3 for thermal impedance graph.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25 °C.

Limits	h_{FE1} (1) V_{CE} = 5 V dc I_C = 10 A dc	h_{FE2} (1) V_{CE} = 5V dc I_C = 20 A dc	$V_{BE(sat)1}$ I_C = 15 A dc I_B = 1.2 A dc	$V_{CE(sat)1}$ I_C = 15 A dc I_B = 1.2 A dc
			V dc	V dc
Min	75	50		
Max	500		2.6	1.5

Limits	$\|h_{fe}\|$	C_{obo}	Pulse response (2)			
	V_{CE} = 10 V dc I_C = 1 A dc f = 1 MHz	V_{CB} = 10 V dc I_E = 0 100 KHz < f < 1 MHz	t_d	t_r	t_s	t_f
		pF	µs	µs	µs	µs
Min	25	150				
Max		375	0.1	0.3	1.2	0.3

(1) Pulsed (see 4.5.1).
(2) See figure 4 for pulse response circuits.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

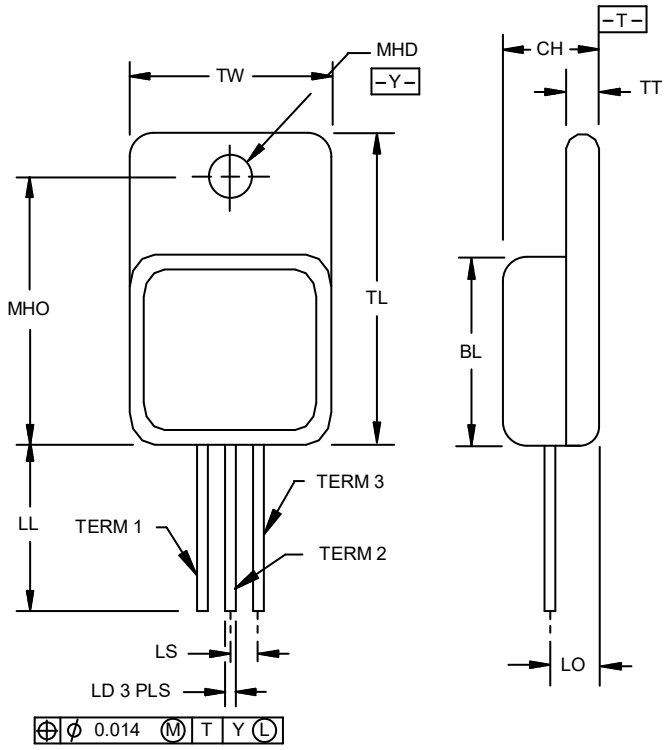
1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are "7575", "7576", and "7577".

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4of this specification, whether or not they are listed.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.95	14.48
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84
Term 1	Emitter			
Term 2	Collector			
Term 3	Base			

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Protrusion of ceramic eyelets included in dimension LL.
3. All terminals shall be isolated from case.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Dimensions and configuration of three terminal flange mount header package (modified TO-254AA).

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Acronyms, symbols, and definitions. The acronyms, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

I_H – The collector current applied to the device under test during the heating period.
I_M – The measurement current applied to forward bias the junction for measurement of V_{BE}.
t_H – The duration of the applied heating power pulse.
t_{SW} – Sample window time during which final V_{BE} measurement is made.

3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Lead finish. The lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Lead formation. Where a choice of lead formation is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform hermetic seal in accordance with screen 14 of table E-IV of MIL-PRF-19500 and 100 percent dc testing in accordance with table I, subgroup 2 herein on all devices that had its leads altered.

3.4.3 Lead isolation. Methods used for electrical isolation of the terminal feedthroughs shall employ material that contain a minimum of 90 percent ceramic AL_2O_3 or equivalent. Examples of such construction techniques are metalized eyelets or ceramic walled packages.

3.4.4 Polarity. The identification of terminals of the device package shall be as shown on figure 1.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (quality levels JANS, JANTXV, and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS	JANTX and JANTXV
3c (1)	Thermal impedance (see 4.3.1)	Thermal impedance (see 4.3.1)
9	I_{CEX1}	I_{CEX1}
11	I_{CEX1} and h_{FE2}, ΔI_{CEX1} = 100 percent of initial value or 50 nA dc, whichever is greater.	I_{CEX1} and h_{FE2}, ΔI_{CEX1} = 100 percent of initial value or 100 nA dc, whichever is greater.
12	See 4.3.2	See 4.3.2
13	Subgroups 2 and 3 of table I herein. ΔI_{CEX1} = 100 percent of initial value or50 nA dc, whichever is greater. Δh_{FE2} = ±15 percent of initial value.	Subgroup 2 of table I herein. ΔI_{CEX1} = 100 percent of initial value or 100 nA dc, whichever is greater. Δh_{FE2} = ±25 percent of initial value.
17	Dielectric withstanding voltage (see 4.3.3).	Dielectric withstanding voltage (see 4.3.3).

(1) Shall be performed anytime after temperature cycling, screen 3a; Quality levels JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with test method 3131 of MIL-STD-750 using the guidelines in that test method for determining I_M, I_H, t_H, t_{SW}, t_{MD} (and V_H where appropriate). The thermal impedance limit used in 4.3, screen 3c, and table I, subgroup 2 herein shall be set statistically by the supplier. See group E inspection (table II, subgroup 4) herein.

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4.3.2 Power burn-in conditions. Power burn-in conditions shall be as follows: T_J = +175 °C minimum, V_{CB} greater than or equal to 25 V dc; T_A = +30 °C maximum.

4.3.3 Dielectric withstanding voltage (for TO-254 packages only). The test for dielectric withstanding voltage shall be performed in accordance with method 1081 of MIL-STD-750. After the test, electrical end-points in accordance with subgroup 2 of table I herein shall be taken. The following details shall apply:

a. Magnitude of test voltage……..900 V dc.

b. Duration of application of test voltage……..15 seconds (min).

c. Points of application of test voltage……...All terminals to case (bunch connection).

d. Method of connection……..Mechanical.

e. Kilovolt-ampere rating of high voltage source…….1,200 V/1.0 mA (min).

f. Maximum leakage current……...1.0 mA.

g. Voltage ramp up time.....500 V/second.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of table I, subgroups 1 and 2. Group B inspection in accordance with subgroup 1 of table E-VIB of MIL-PRF-19500 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. Electrical end-point measurements shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in 4.4.2.1 for quality level JANS and and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV. Delta requirements shall be in accordance with the applicable step of 4.6 herein.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

Subgroup	Method	Conditions
B3	1037	For solder die attach: V_{CB} greater than or equal to 100 V dc.
B4	1027	For eutectic die attach: V_{CB} greater than or equal to 100 V dc; adjust P_T to achieve T_J = +175 °C minimum; T_A = +30 °C maximum.

4.4.2.2 Quality level levels JAN, JANTX, and JANTXV (table E-VIB of MIL-PRF-19500).

Subgroup	Method	Conditions
B3	1027	For eutectic die attach: V_{CB} greater than or equal to 100 V dc; adjust P_T to achieve T_J = +175 °C minimum; T_A = +30 °C maximum.
B3	1037	For solder die attach: V_{CB} greater than or equal to 100 V dc.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows herein. Delta measurements shall be in accordance with the applicable step of 4.6 herein.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 10 pounds (4.54 Kg); time = 15 seconds.
C5	3131	See 4.3.1, $R_{\theta JC}$ = 1.0 °C/W.
C6	1027	For eutectic die attach: V_{CB} greater than or equal to 100 V dc; adjust P_T to achieve T_J = +175 °C minimum; T_A = +30 °C maximum, 1,000 hours minimum.
C6	1037	For solder die attach: V_{CB} greater than or equal to 100 V dc.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Delta measurements shall be in accordance with the applicable step of 4.6 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse response measurements. The conditions for pulse response measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Insulation resistance test. The isolation resistance test shall be in accordance with test method 1016 of MIL-STD-750. The test voltage shall be 1000 volts. The following preparations shall apply: short the collector, emitter and base terminals together. The limit for acceptance shall be 10^9 Ω minimum.

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4.6 Delta measurements. The requirements for delta measurements for groups B, C, and E shall be as specified below.

Steps	Inspection (1) (2) (3) (4) (5)	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to base cutoff current 2N7575 2N7576 2N7577	3036	Bias condition D V_{CB} = 350 V dc V_{CB} = 400 V dc V_{CB} = 450 V dc	ΔI_{CBO}	100 percent of initial value or 100 nA, whichever is greater.		
2	Forward − current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 20 A dc; pulsed (see 4.5.1)	Δh_{FE2}	±25 percent change from initial reading.		
3	Saturation voltage, collector to emitter	3071	I_C = 15 A dc; I_B = 1.2 A dc, pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$	±200 mV change from previously measured value.		

(1) Devices which exceed the group A limits for this test shall not be acceptable.
(2) The delta electrical measurements for group B, product assurance level JANS, shall be as follows:
   a. In addition to the measurements specified for subgroup 4 of table E-VIA of MIL-PRF-19500, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 5 of table E-VIA of MIL-PRF-19500, the measurements of steps 1 and 2 of this table shall also be taken.
(3) The delta electrical measurements for group B, product assurance levels JAN, JANTX and JANTXV, shall be as follows:
   a. In addition to the measurements specified for subgroup 3 of table E-VIB of MIL-PRF-19500, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6 of table E-VIB of MIL-PRF-19500, the measurements of step 2 of this table shall also be taken.
(4) The delta electrical measurements for group C, all product assurance levels, shall be as follows:
   a. In addition to the measurements specified for subgroup 2 of table E-VII of MIL-PRF-19500, the measurements of step 1 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 3 of table E-VII of MIL-PRF-19500, the measurements of step 1 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 6 of table E-VII of MIL-PRF-19500, the measurements of steps 1, 2, and 3 of this table shall also be taken.
(5) The delta measurements for group E, all product assurance levels, shall be as follows: In addition to the measurements specified for subgroups 1 and 2 of table E-IX of MIL-PRF-19500, the measurements of steps 1, 2, and 3 of this table shall also be taken.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.1	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N7575				300		V dc
2N7576				350		V dc
2N7577				400		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = −1.5 V dc	I_{CEX1}			
2N7575		V_{CE} = 350 V dc			0.1	mA dc
2N7576		V_{CE} = 400 V dc			0.1	mA dc
2N7577		V_{CE} = 450 V dc			0.1	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = −1.5 V dc	I_{CEO}			
2N7575		V_{CE} = 300 V dc			0.1	mA dc
2N7576		V_{CE} = 350 V dc			0.1	mA dc
2N7577		V_{CE} = 400 V dc			0.1	mA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N7575		V_{CBO} = 350 V dc			0.1	mA dc
2N7576		V_{CBO} = 400 V dc			0.1	mA dc
2N7577		V_{CBO} = 450 V dc			0.1	mA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc	I_{EBO}		300	mA dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 15 A dc; pulsed (see 4.5.1); I_B = 1.2 A dc	$V_{BE(SAT)}$		2.6	V dc
Saturation voltage; collector to emitter	3071	I_C = 15 A dc; pulsed (see 4.5.1) I_B = 1.2 A dc	$V_{CE(SAT)1}$		1.5	V dc
Saturation voltage; collector to emitter	3071	I_C = 20 A dc; pulsed (see 4.5.1) I_B = 1.2 A dc	$V_{CE(SAT)2}$		1.7	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE1}	75	500	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 20 A dc; pulsed (see 4.5.1)	h_{FE2}	50		

See footnotes at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150 °C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = −1.5 V dc	I_{CEX2}			
2N7575		V_{CE} = 350 V dc			0.1	mA dc
2N7576		V_{CE} = 400 V dc			0.1	mA dc
2N7577		V_{CE} = 400 V dc			0.1	mA dc
Low-temperature operation:		T_A = −55 °C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE3}	20		
Subgroup 4						
Magnitude of common emitter small-signal short-circuit forward- current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1 A dc; f = 1 MHz	$\|h_{fe}\|$	25		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1.0 MHz	C_{obo}	150	375	pF
Pulse response:	3251	Test condition A except test circuit and pulse requirements in accordance with figure 4, T_A = +25 °C, V_{CC} = 300V, I_C = 10A, I_{B1} = 0.5A, $V_{BE(off)}$ = −5V, tp = 50 µs				
Pulse delay time			t_d		0.1	µs
Pulse rise time			t_r		0.3	µs
Pulse storage time			t_s		1.2	µs
Pulse fall time			t_f		0.3	µs
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25 °C; t = 1 s; 1 cycle; see figure 5				
Test 1						
(All device types)		V_{CE} = 8.0 V dc; I_C = 20 A dc				

See footnotes at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 – Continued.						
Test 2						
(All device types)		V_{CE} = 30 V dc; I_C = 3.0 A dc				
Test 3						
(All device types)		V_{CE} = 100 V dc; I_C = 0.17 A dc				
Test 4						
2N7575		V_{CE} = 300 V dc; I_C = 25 mA dc				
2N7576		V_{CE} = 350 V dc; I_C = 19 mA dc				
2N7577		V_{CE} = 400 V dc; I_C = 15 mA dc				
Safe operating area (switching); inductive collector to emitter sustaining voltage	3053	T_A = +25 °C, I_C = 2 A, I_{B1} = 0.1 A, I_{B2} = 0 A, L = 1 mH, clamped at rated V_{CEO}	$V_{CEO(sus)}$			
2N7575		Clamp voltage = 300 V dc		300		V dc
2N7576		Clamp voltage = 350 V dc		350		V dc
2N7577		Clamp voltage = 400 V dc		400		V dc
Safe operating area (switching); inductive collector to emitter sustaining voltage	3053	T_A = +25 °C, I_C = 2 A, I_{B1} = 0.5 A, $V_{BE(off)}$ = 1V, L = 1 mH, clamped at rated V_{CEO}	$V_{CEX(sus)}$			
2N7575		Clamp voltage = 300 V dc		300		V dc
2N7576		Clamp voltage = 350 V dc		350		V dc
2N7577		Clamp voltage = 400 V dc		400		V dc
Subgroup 6						
Not applicable						
Subgroup 7						
Insulation resistance test	1016	See 4.5.2.	R_{ISO}	10^9		Ω

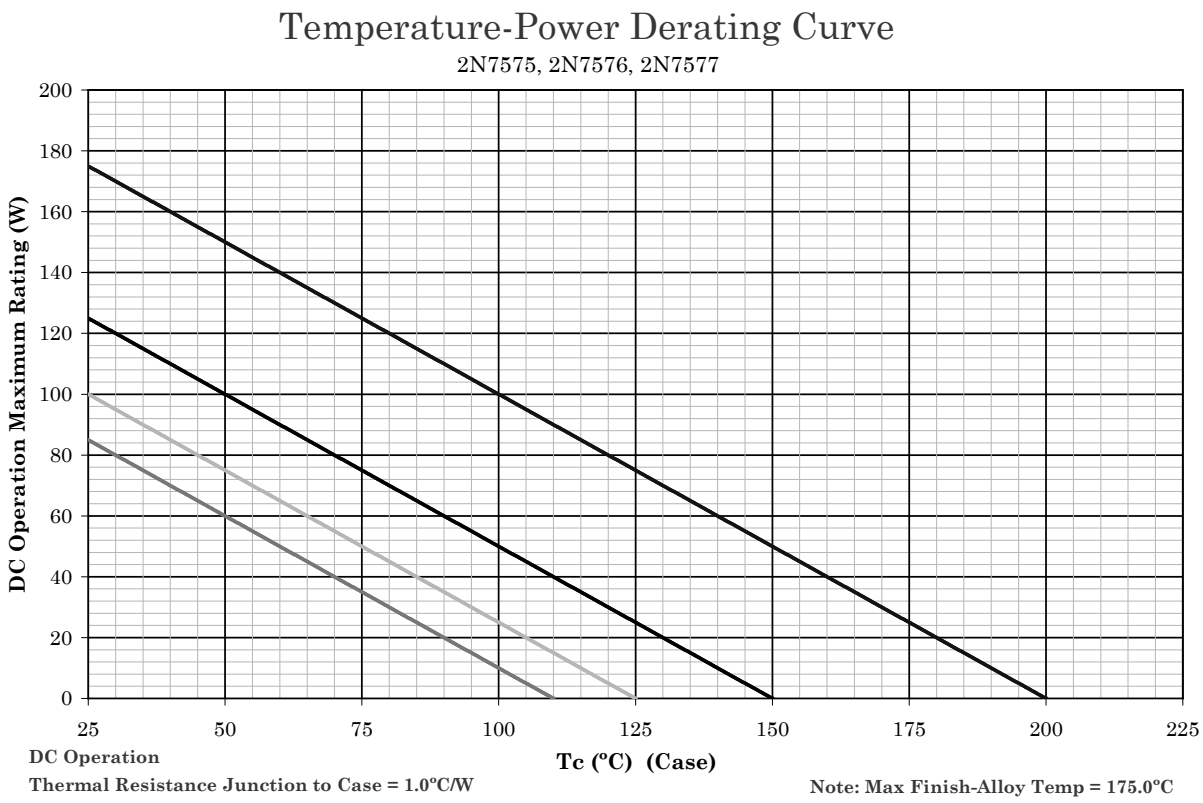
1/ For sampling plan see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) − for qualification and re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock (liquid to liquid)	1056	0°C to +100°C, 100 cycles	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 2			45 devices, c = 0
Blocking life	1048	1,000 hours minimum, T_A = +150°C, V_{CB} = 80 percent of rated.	
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			15 devices, c = 0
Barometric pressure (reduced)	1001	Condition C. See 1.3.	
Subgroup 6			
Electrostatic discharge sensitivity	1020		
Subgroup 8			45 devices, c = 0
Reverse voltage leakage stability	1033	Condition A for all devices.	

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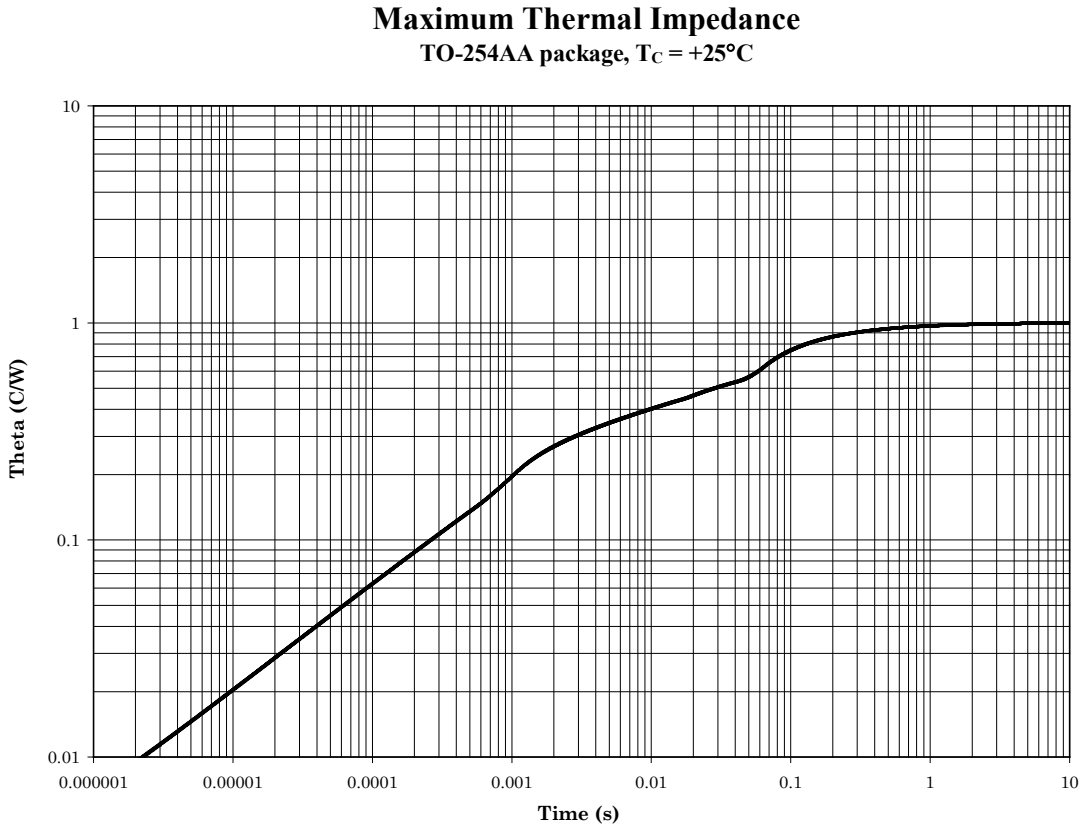


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +200 °C) and power rating specified (see 1.3 herein).
3. Derate design curve chosen at T_J ≤ +150 °C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ +125 °C, and +110 °C to show power rating where most users want to limit T_J in their application.

FIGURE 2. Temperature-power derating graph.

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$R_{\theta JC}$ = 1.0 °C/W max.

FIGURE 3. Thermal impedance graph.

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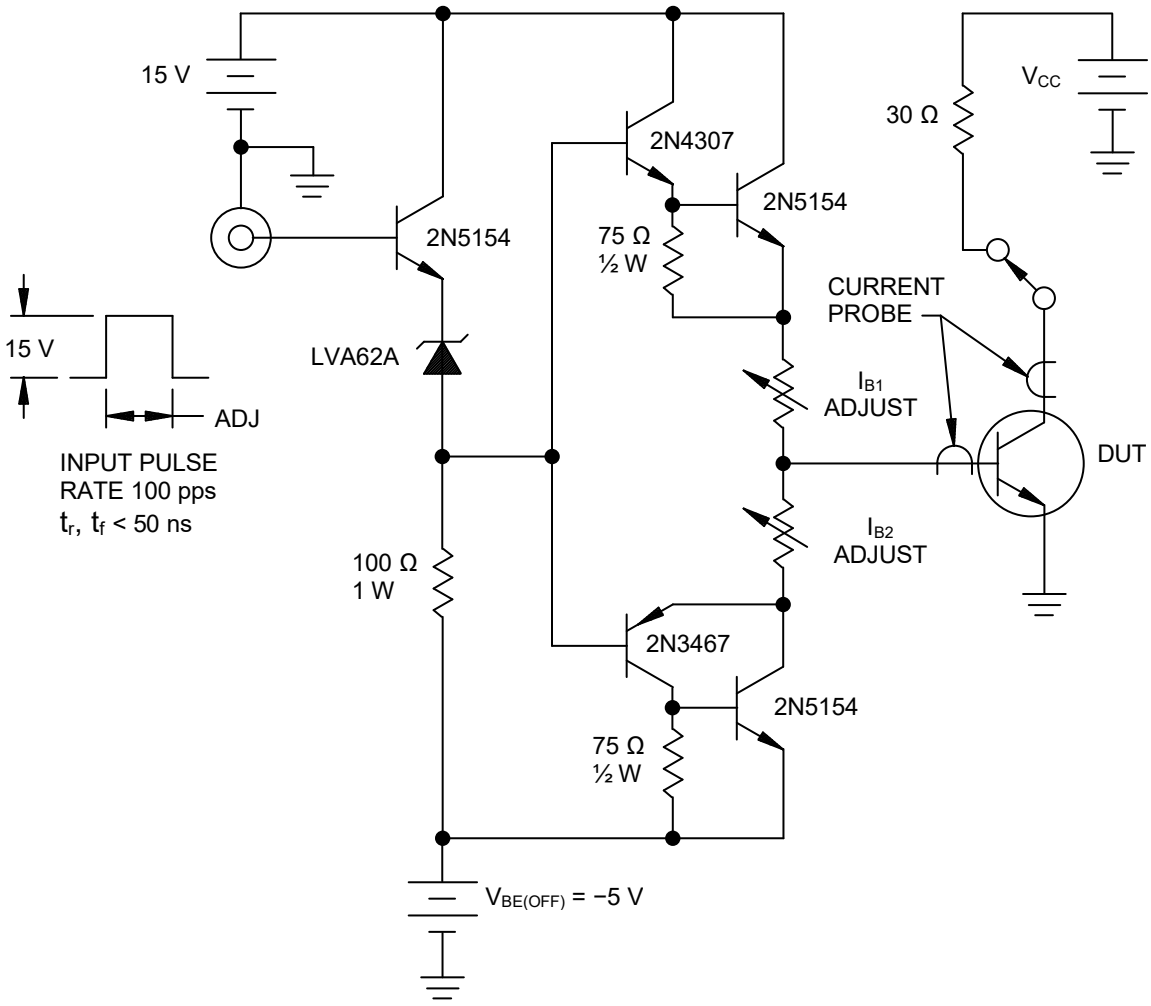


FIGURE 4. Pulse response test circuit.

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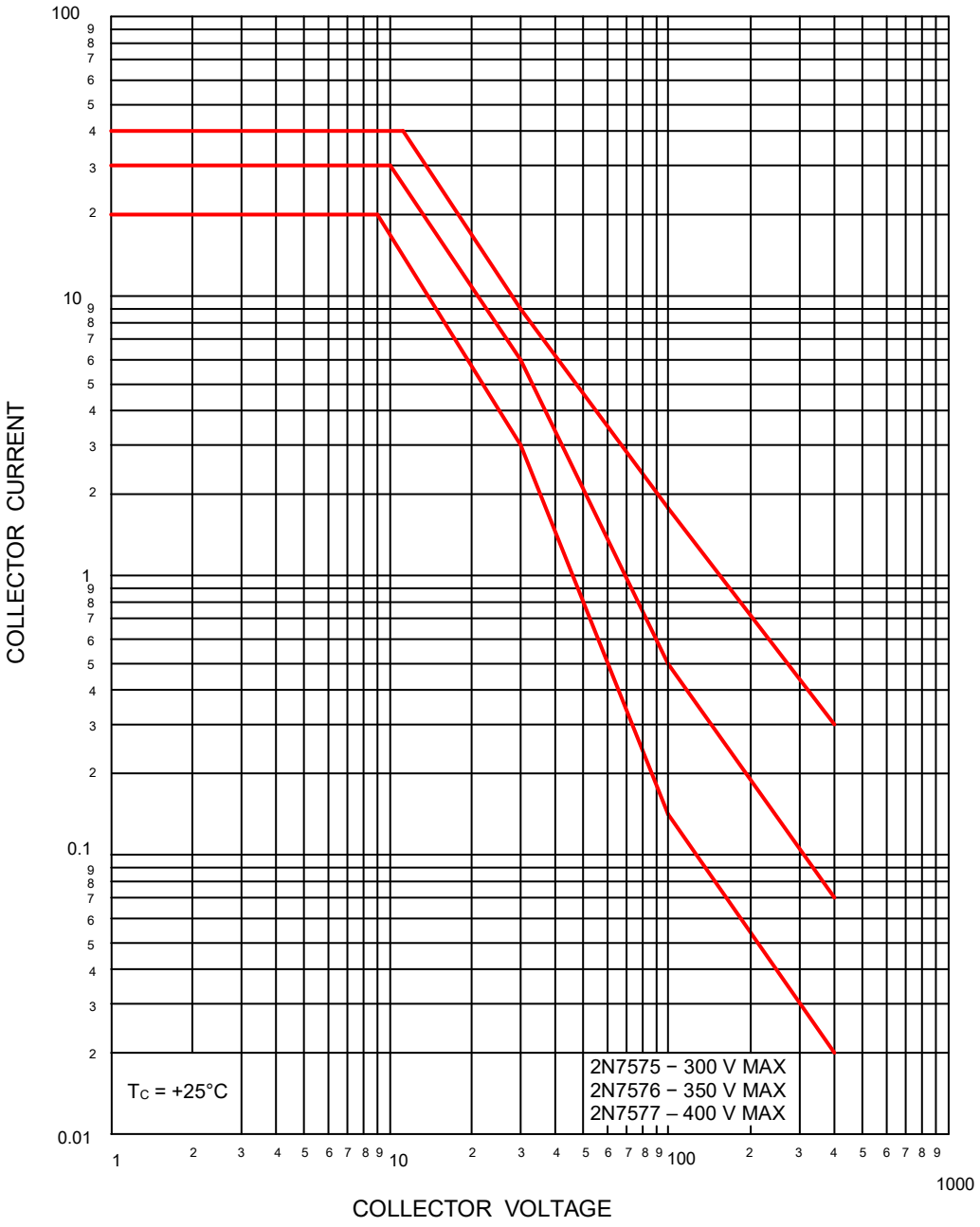


FIGURE 5. Maximum safe operating area graph.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

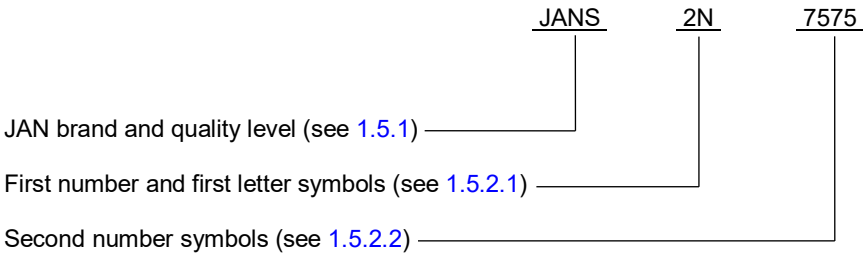
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1) and if applicable, any needed lead formation (see 3.4.2).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form:



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6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

Base quality level	Quality level "TX"	Quality level "TXV"	Quality level "S"
JAN2N7575	JANTX2N7575	JANTXV2N7575	JANS2N7575
JAN2N7576	JANTX2N7576	JANTXV2N7576	JANS2N7576
JAN2N7577	JANTX2N7577	JANTXV2N7577	JANS2N7577

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army − CR
Navy − EC
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − MI
Air Force − 19

Preparing activity:
DLA − CC

(Project 5961−2020−031)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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